

Varshini Elangovan

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EDUCATION

Carnegie Mellon University

Pittsburgh, PA

Master of Science in Machine Learning

Expected: December 2026

- *Relevant Coursework:* Optimization for Machine Learning*, Probabilistic Graphical Models*, Advanced Introduction to Machine Learning, Deep Reinforcement Learning, Probability and Mathematical Statistics
- *Research Interests:* Multimodal Reasoning; Reinforcement Learning; Interpretable, Reliable and Safe AI

Anna University, College of Engineering, Guindy

Chennai, India

Bachelor of Engineering (Honors) in Computer Science and Engineering

June 2022

WORK EXPERIENCE

Google (Global Payments Platform Risk)

Hyderabad, India

Software Engineer III

November 2024 - July 2025

Software Engineer II

July 2022 - October 2024

- Primary contributor to the risk strategy design for Google Wallet via PIX in Brazil launched in August 2024. Developed risk mitigation solutions to reduce transaction losses across peer-to-peer payment flows.
- Designed and deployed feature engineering pipelines in Java and rule-based decision policies in Python, integrating ML model outputs for online transaction risk assessment at scale.
- Implemented risk capabilities compliant with regulatory requirements including real-time monitoring dashboards and operational runbooks. Collaborated with Trust & Safety and Product teams on metrics, thresholds and customer impact post launch.
- Engineered ML pipelines for delinquency fraud detection on Stored Value transactions, performing data simulation, GBDT model training in TensorFlow and offline evaluation against baselines.

India Urban Data Exchange (IUDX)

May 2022 - June 2022

Intern

Remote (Bangalore, India)

- Evaluated Facebook AI Similarity Search (FAISS) as a potential alternative to Elasticsearch for improving dataset indexing on the IUDX Dataset Catalogue. Devised a framework to enhance search accuracy with the integration of Google's T5 model for NLP-to-SQL translation.

Goldman Sachs

June 2021 - July 2021

Summer Analyst

Remote (Bangalore, India)

- Created a Python-based data curation and analysis tool that converts JSON inputs into multiple outputs for GS proprietary systems. Integrated the same into ETL workflows, optimizing a 2-week process to under one minute.

RESEARCH EXPERIENCE

Human Sensing Lab, Carnegie Mellon University

October 2025 - Present

Graduate Student Researcher

Pittsburgh, PA

- Working on 3D Spatial Understanding with 2D Vision-Language models under the guidance of Prof. Fernando De la Torre, Robotics Institute in collaboration with Meta Reality Labs.
- Developing 3D-guided, multi-view adaptive visual token pruning methods to reduce cross view redundancy and select question-conditioned visual evidence for 3D visual question answering in indoor scene settings.

Indian Institute of Technology (IIT), Kharagpur

May 2022 - September 2022

Deep Learning Intern

Remote (Kharagpur, India)

- Collaborated on a Depth Estimation task to predict accurate depth from simulated colonoscopy images using deep learning models at the Kharagpur Learning, Imaging, and Visualization (KLIV) research group.
- Trained and evaluated semantic segmentation models in PyTorch for depth estimation from simulated colonoscopy images as part of the KLIV group, assessing various loss functions and effective post-processing techniques. Presented at the MICCAI 2022 Endoscopic Vision Challenge.

- Analyzed the applications of deep learning in solving inverse scattering problems at Signal Processing Interpretation and Representation (SPIRE) Lab by employing deep neural networks to model the estimation of scatterer properties from arbitrary sensor locations.
- Formulated the problem statement of reconstructing a global scattered field from a sparse sensor configuration, studied and implemented various research papers in Tensorflow and investigated results in MATLAB.

PROJECTS

Watch & Learn: Teacher-Student Distillation for Robotic Arm Tasks

PyTorch, OpenAI Gym | Reinforcement Learning | [GitHub](#)

- Developed a vision-only agent for robotic manipulation under partial observability, using teacher-guided RL approach combining imitation and critic-based feedback to achieve 90% of expert performance while improving sample efficiency.

Environmental ML: Modelling Sequential Disaster Cascades

XGBoost, Scikit-learn, PyTorch | Multi-label Classification

- Built an end-to-end ML pipeline on 15 years of NOAA Storm Events data to model multi-hazard disaster cascades under extreme class imbalance. Designed weather-embedded neural nets focusing on rare event detection and robustness across imbalanced labels.

Let's Dance: Music-conditioned 2D Dance Movement Generation

PyTorch, OpenPose | Generative Deep Learning | [Report](#)

- Implemented an end-to-end GAN framework in PyTorch to synthesize dance motion videos based on audio, leveraging OpenPose 2D skeletal key points and a pix2pix model to generate photo-realistic motion sequences.

Adaptive Day Planner: LLM-Based Task Scheduling and Voice Agent

Mistral LLMs, FastAPI | Agentic AI | [GitHub](#)

- Built an AI-powered adaptive planner using Mistral LLMs, FastAPI and Eleven Labs speech APIs to dynamically schedule tasks, replan sessions based on user feedback and support focus sessions via conversational body doubling. (Mistral AI Worldwide Hackathon, SF)

Reverse Image Search Engine for Garment Industry

TensorFlow, Keras, Flask | Content-Based Image Retrieval | [GitHub](#)

- Developed a reverse image search engine to retrieve visually similar images from a fashion database via an user interface with feature vectors extracted using ResNet-50. Optimized feature similarity matching using Spotify's Annoy Indexing algorithm. Co-first authored and presented a paper at an IEEE Conference.

PUBLICATIONS

- SimCol3D – 3D reconstruction during colonoscopy challenge, A. Rau, S. Bano, Y. Jin, Varshini Elangovan et al., *Medical Image Analysis*, 2024. Reference: [doi: 10.1016/j.media.2024.103195](#)
- Reverse Image Search Engine for Garment Industry, Anushri Eswaran and Varshini Elangovan, *8th International Conference on Advanced Computing and Communication Systems (ICACCS)*, 2022. Reference: [doi: 10.1109/ICACCS54159.2022.9785089](#)

SKILLS

Languages: Python, Java, C/C++, SQL, JavaScript

Frameworks/Libraries: PyTorch, TensorFlow, Keras, NumPy, Pandas, scikit-learn, Hugging Face, OpenCV, Guice

Developer Tools: Git, Docker, Weights & Biases, FastAPI, MATLAB

COURSES

BlueDot Impact Technical AI Safety

April 2026 (Ongoing)